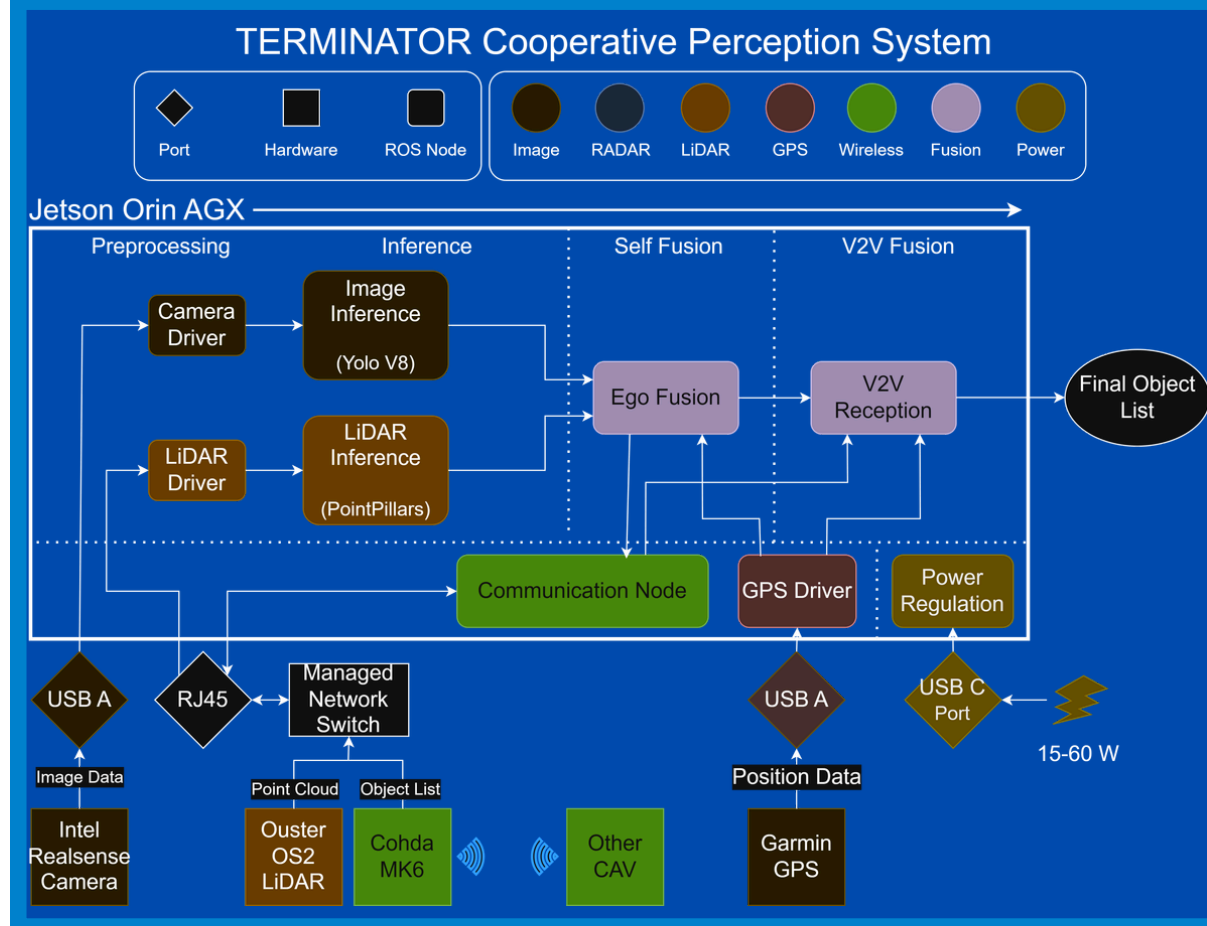
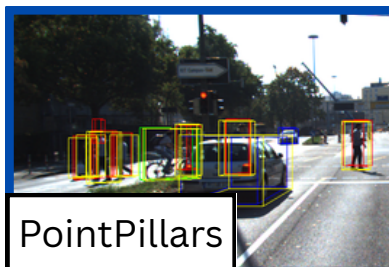


TERMINATOR: Truck Enabled Reconnaissance via Multi-modal Integration for Networked Autonomous Traffic Observation and Response

Hardware & Software

The TERMINATOR system runs on ROS Noetic on an NVIDIA Jetson AGX Orin, using modular ROS nodes for data acquisition, preprocessing, inference, fusion, and communication. An Intel RealSense D455 depth camera, Ouster OS2 LiDAR, and GNSS provide sensor data. YOLOv8 performs real-time object detection, on camera frames while PointPillars processes 3D point clouds. Final object lists are transmitted via LTE-V2X in ETSI message format.



Abstract

Advances in artificial intelligence inference hardware have made real-time, local, multi-modal object detection increasingly practical. Combined with the growing availability of Cellular Vehicle-to-Everything (C-V2X) communication technologies, this has driven significant interest in cooperative perception systems that allow vehicles to share sensor information and extend awareness beyond their own field of view. However, most existing research has been conducted in simulation environments such as CARLA, with limited opportunities for physical testing and validation.

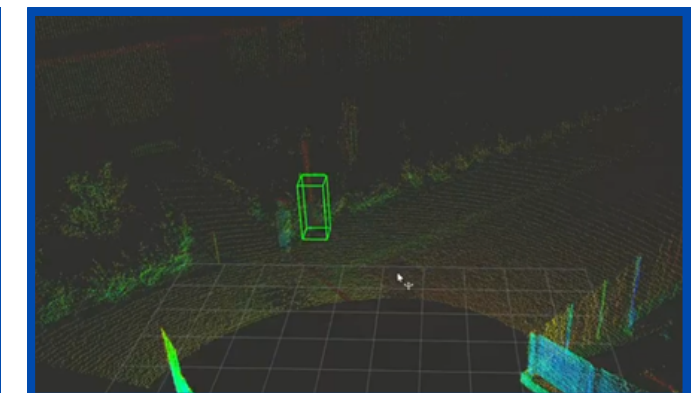
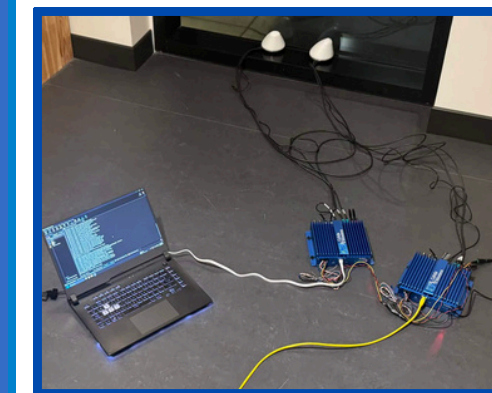
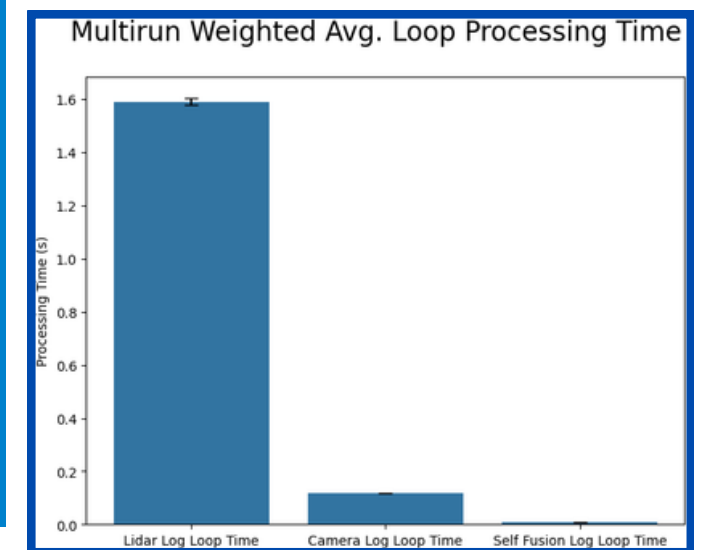
TERMINATOR addresses this gap by developing a small-scale, physically testable cooperative perception platform capable of near real-time object detection, object-list fusion, and inter-vehicle data transmission. The system is built around an NVIDIA Jetson AGX Orin and utilizes camera, LiDAR, and GNSS data to generate and share environmental information. ROS Noetic provides modular communication between system components, enabling sensing, perception, and data exchange. By supporting cooperative perception experiments in a controlled physical environment, TERMINATOR provides an accessible platform for evaluating real-world multi-modal perception and communication techniques.

Results

The TERMINATOR successfully demonstrated a bench-top cooperative perception system capable of performing camera and LiDAR inference, multimodal object-list fusion, and object-list transmission between Cohda MK6 devices. The system also achieved functional GNSS-based localization while maintaining a compact form suitable for laboratory testing and experimentation.

Performance measurements showed that the camera inference and fusion pipelines operated within fractions of a second, enabling near real-time processing. However, LiDAR inference required several seconds per frame, representing the primary bottleneck in system performance. Future improvements may include point cloud cropping, reduced point cloud resolution, or the adoption of more computationally efficient detection algorithms to decrease processing time.

Overall, the project provides a successful proof of concept for a physically testable cooperative perception platform. While additional optimization is required to achieve fully real-time operation, the results demonstrate the feasibility of integrating multimodal perception, localization, and vehicle-to-vehicle communication within a compact experimental system.



References

- [1] Z. Bai, G. Wu, and M. Barth, "A Survey and Framework of Cooperative Perception: From Heterogeneous Singleton to Hierarchical Cooperation," IEEE Trans. Intelligent Transportation Systems, Aug. 2022.
- [2] A. Geiger, P. Lenz, and R. Urtasun, "Are We Ready for Autonomous Driving? The KITTI Vision Benchmark Suite," 2012.
- [3] Y. Cohen, A. Biton, and S. Shoval, "Fusion of Computer Vision and AI in Collaborative Robotics: A Review and Future Prospects," Applied Sciences, vol. 15, no. 14, Jul. 2025.
- [4] P. Jiang et al., "A Review of YOLO Algorithm Developments," Procedia Computer Science, vol. 199, Jan. 2022.
- [5] W. Liu et al., "SSD: Single Shot MultiBox Detector," Dec. 2015.
- [6] H. Wu et al., "Virtual Sparse Convolution for Multimodal 3D Object Detection," CVPR, 2023.
- [7] A. H. Lang et al., "PointPillars: Fast Encoders for Object Detection from Point Clouds," 2019.
- [8] R. Nabati and H. Qi, "CenterFusion: Center-Based Radar and Camera Fusion for 3D Object Detection," 2021.
- [9] NVIDIA, "Jetson AGX Orin Developer Kit User Guide," 2024.
- [10] Wikipedia Contributors, "Z-order Curve," 2019.

ROS Nodes	
NODE	INPUTS
PointPillars	LIDAR PointCloud
YOLOv8	Camera Image
Fusion	LIDAR Inference Camera Inference
ETSI Comms	Fused Object List